

Arogya – Technology Stack Document

Project Name: Arogya – AI-Powered Holistic Health Super App

Version: 1.0

Document Type: Technology Stack Documentation

1. Overview

Arogya is a modern full-stack web application built using a scalable JavaScript ecosystem. The technology stack is selected to ensure high performance, rapid development, security, and seamless deployment while supporting future expansion such as AI services, telehealth integration, and wearable device connectivity.

2. Technology Stack Overview

Layer	Technology
Frontend	React.js
Styling	Tailwind CSS
UI Components	shadcn/ui
Icons	Lucide React
Animations	Framer Motion
State Management	Context API / Redux Toolkit
Routing	React Router DOM
Backend	Supabase
Database	PostgreSQL
Authentication	Supabase Auth
Storage	Supabase Storage
Real-time	Supabase Realtime
Server Logic	Supabase Edge Functions
Charts	Chart.js / Recharts
Form Validation	React Hook Form + Zod
HTTP Client	Supabase SDK / Fetch API

Layer	Technology
Notifications	React Hot Toast
AI Integration	OpenAI API (future)
Voice Features	Web Speech API
Version Control	Git & GitHub
Deployment	Vercel
Testing	Vitest + React Testing Library

3. Frontend Technologies

React.js

Purpose

Develop a modern Single Page Application (SPA).

Why React?

- Component-based architecture
- Fast rendering with Virtual DOM
- Large ecosystem
- Easy state management
- Highly scalable
- Excellent community support

Usage

- Dashboard
 - Health Tracking
 - Reports
 - Authentication
 - AI Chat Interface
 - User Profile
 - Analytics
-

React Router DOM

Purpose

Client-side routing.

Routes

- Landing Page
 - Login
 - Register
 - Dashboard
 - Nutrition
 - Fitness
 - Mental Health
 - Women's Health
 - AI Coach
 - Reports
 - Settings
-

Tailwind CSS

Purpose

Responsive UI development.

Advantages

- Utility-first CSS
- Mobile-first design
- Faster development
- Consistent styling
- Easy customization

Used for

- Layout
 - Cards
 - Forms
 - Dashboard
 - Navigation
 - Dark Mode
-

shadcn/ui

Purpose

Accessible and reusable UI components.

Components

- Buttons
 - Cards
 - Dialogs
 - Forms
 - Tabs
 - Modals
 - Tables
 - Toasts
-

Framer Motion

Purpose

Modern animations.

Examples

- Dashboard transitions
 - Card animations
 - Progress indicators
 - AI avatar animations
 - Page transitions
-

4. Backend Technologies

Supabase

Supabase provides a Backend-as-a-Service (BaaS) platform, eliminating the need for a traditional backend server.

Services Used

- Authentication
- PostgreSQL Database
- Storage

- Edge Functions
- Realtime
- Row Level Security (RLS)

Why Supabase?

- Open source
 - PostgreSQL support
 - Real-time database
 - Easy authentication
 - Built-in APIs
 - Serverless architecture
 - Excellent React integration
-

5. Database

PostgreSQL

Purpose

Store structured healthcare information.

Main Tables

- users
- profiles
- health_metrics
- nutrition_logs
- workout_logs
- sleep_logs
- mood_logs
- period_tracker
- pregnancy_tracker
- hormone_tracker
- achievements
- badges
- challenges
- notifications
- ai_conversations
- appointments
- doctors
- community_posts

Features

- ACID compliance

- Relationships
 - Indexing
 - JSON support
 - High reliability
-

6. Authentication

Supabase Auth

Supports

- Email & Password
- Google Login
- GitHub Login
- Password Reset
- Session Management

Security Features

- JWT Tokens
 - Secure Cookies
 - Email Verification
 - Multi-provider Authentication
-

7. Storage

Supabase Storage

Stores

- Profile Photos
- Medical Reports
- Health Documents
- AI-generated Images (future)

Advantages

- Secure file access
 - Public/Private buckets
 - Easy integration
-

8. State Management

Context API

Used for

- Authentication State
- Theme Management
- User Settings

Redux Toolkit (Optional)

Used for

- Dashboard data
 - Health metrics
 - Notifications
 - Global application state
-

9. Data Visualization

Chart.js / Recharts

Used for

- Weight Progress
 - Sleep Trends
 - Mood Analytics
 - Calories
 - Water Intake
 - Heart Rate
 - Weekly Reports
-

10. Form Management

React Hook Form

Features

- Fast validation
- Minimal re-rendering

- Easy integration

Zod

Used for

- Schema validation
 - Form validation
 - API validation
-

11. AI Integration

OpenAI API (Future Enhancement)

Purpose

Power the AI Health Coach.

Capabilities

- Personalized health guidance
 - Nutrition recommendations
 - Habit suggestions
 - Daily motivation
 - Health diary summaries
 - Conversational assistance
-

12. Voice Features

Web Speech API

Features

- Voice navigation
 - Voice commands
 - Speech-to-text
 - Text-to-speech
 - Accessibility support
-

13. JavaScript Algorithms

The application uses custom JavaScript algorithms to generate personalized insights.

Algorithms

Health Score

Inputs

- BMI
- Exercise
- Sleep
- Water Intake
- Mood
- Nutrition

Output

Overall health score (0–100)

Calorie Calculator

Uses

- BMR
 - TDEE
 - Activity Level
 - Weight Goal
-

BMI Calculator

Formula

$BMI = \text{Weight} / \text{Height}^2$

Workout Recommendation

Based on

- Age
- Gender
- Fitness Goal

- BMI
 - Activity Level
-

Water Intake Recommendation

Based on

- Weight
 - Temperature
 - Activity Level
-

Mood Analysis

Based on

- Daily mood logs
 - Sleep quality
 - Stress level
 - Exercise frequency
-

Period Prediction

Based on

- Previous cycles
 - Average cycle length
 - Period history
-

Gamification Engine

Calculates

- XP
 - Levels
 - Badges
 - Streaks
 - Achievements
-

14. Security

Authentication

- JWT Tokens
- Email Verification
- Secure Sessions

Database

- Row Level Security (RLS)
- Parameterized Queries
- Access Policies

Application

- HTTPS
 - Input Validation
 - XSS Protection
 - CSRF Protection
 - Environment Variables
-

15. Version Control

Git

Purpose

- Source control
- Branch management
- Collaboration

Branch Strategy

- main
 - development
 - feature/*
 - bugfix/*
-

GitHub

Purpose

- Repository hosting
 - Pull Requests
 - Code Reviews
 - Issue Tracking
 - CI/CD Integration
-

16. Deployment

Vercel

Purpose

Host the React application.

Features

- Automatic deployments
- GitHub integration
- HTTPS
- Edge Network
- Environment Variables
- Preview deployments

Deployment Flow

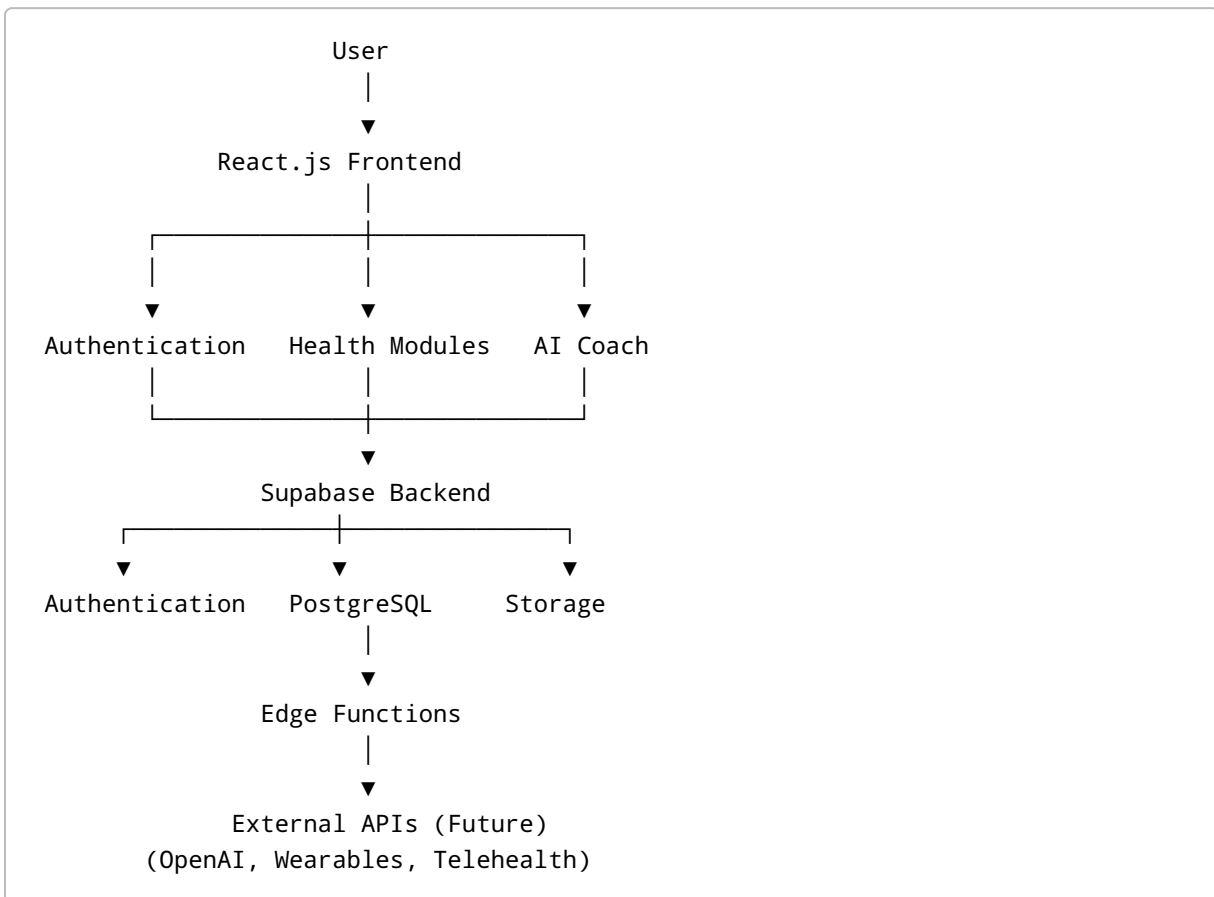
Developer → GitHub → Vercel → Production

17. Development Tools

Tool	Purpose
Visual Studio Code	Code Editor
Git	Version Control
GitHub	Repository Hosting
npm	Package Manager
ESLint	Code Quality

Tool	Purpose
Prettier	Code Formatting
Postman	API Testing
Figma	UI/UX Design

18. Project Architecture



19. Folder Structure

```

src/
├── assets/
├── components/
│   ├── common/
│   └── dashboard/

```

```
|   |— ai/
|   |— health/
|   |— women/
|— pages/
|— layouts/
|— hooks/
|— context/
|— store/
|— services/
|— lib/
|— utils/
|— algorithms/
|— types/
|— styles/
|— App.jsx
```

20. Future Technology Enhancements

- Progressive Web App (PWA)
- Offline Support
- Wearable Device Integration (Google Fit, Apple Health)
- AI Vision for Skin and Food Analysis
- WebRTC for Teleconsultation
- Push Notifications
- Multi-language Support
- Predictive Analytics using Machine Learning
- 3D Health Galaxy Visualization with Three.js

21. Conclusion

The selected technology stack enables Arogya to be a secure, scalable, and maintainable health platform. React.js provides a rich user experience, Supabase simplifies backend development with authentication and PostgreSQL, Tailwind CSS and shadcn/ui ensure a modern interface, while Vercel and GitHub streamline deployment and collaboration. This architecture also leaves room for future AI capabilities, telehealth services, wearable integrations, and advanced analytics without major redesign.